Statistical Business Analysis

Week 7 — Introduction to Statistics for Data Science

Project Overview

This project analyzes business sales data using introductory statistical methods. The analysis covers descriptive statistics, data distribution, correlation, hypothesis testing, confidence intervals, and linear regression.

Objectives

Calculate descriptive statistics; examine distributions; test business hypotheses; measure relationships among sales, marketing spend and customers; estimate a 95% confidence interval; and build a regression model.

Setup and Installation

Install Python 3.13+, create a virtual environment, install the packages listed in requirements.txt, and open statistical_analysis.ipynb in Jupyter Notebook or VS Code.

Dataset

Rows	Columns	Main Metrics
100	5	Sales, Marketing Spend, Customers

Descriptive Statistics

Metric	Mean	Median	Std. Dev.	Mode
Sales	55,257.63	54,891.64	6,987.85	39,160.45
Marketing_Spend	6,496.43	6,616.70	1,364.21	3,518.65
Customers	221.47	223.00	47.22	243.00

Hypothesis Testing

1. One-sample t-test: average sales vs. $45,000 benchmark. t = 14.6792, p = 0.000000 → SIGNIFICANT.

2. Independent t-test: North vs. South sales. t = 3.6422, p = 0.000569 → SIGNIFICANT.

3. One-way ANOVA: North vs. South vs. West. F = 8.5532, p = 0.000379 → SIGNIFICANT.

Correlation Analysis

Sales–Marketing correlation = 0.8046. Sales–Customers correlation = 0.3763. The correlation values describe the strength and direction of linear relationships.

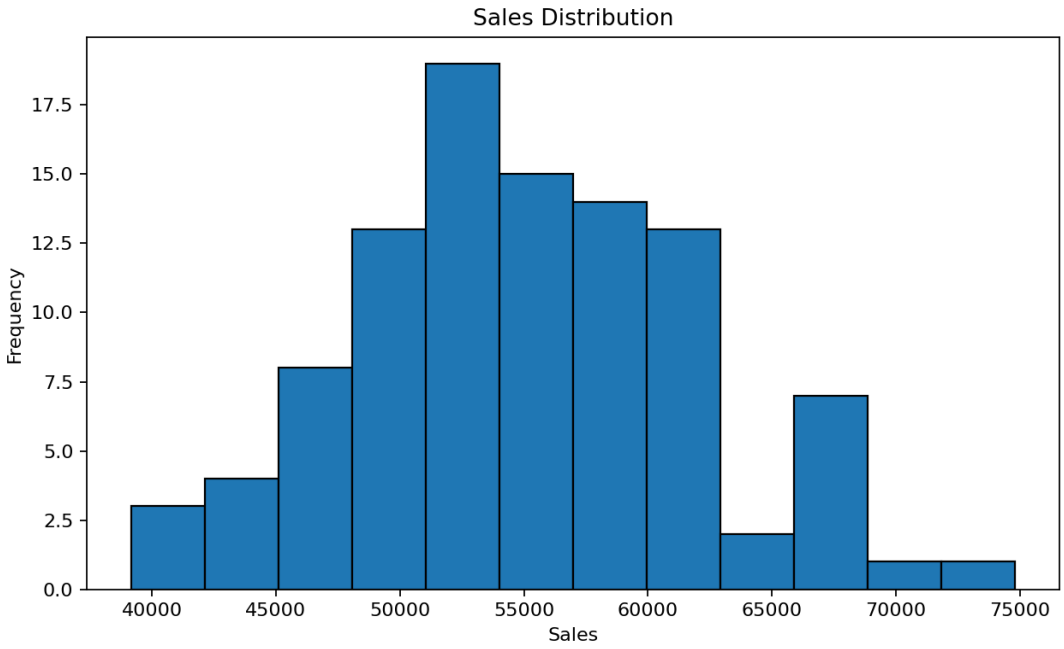
Confidence Interval

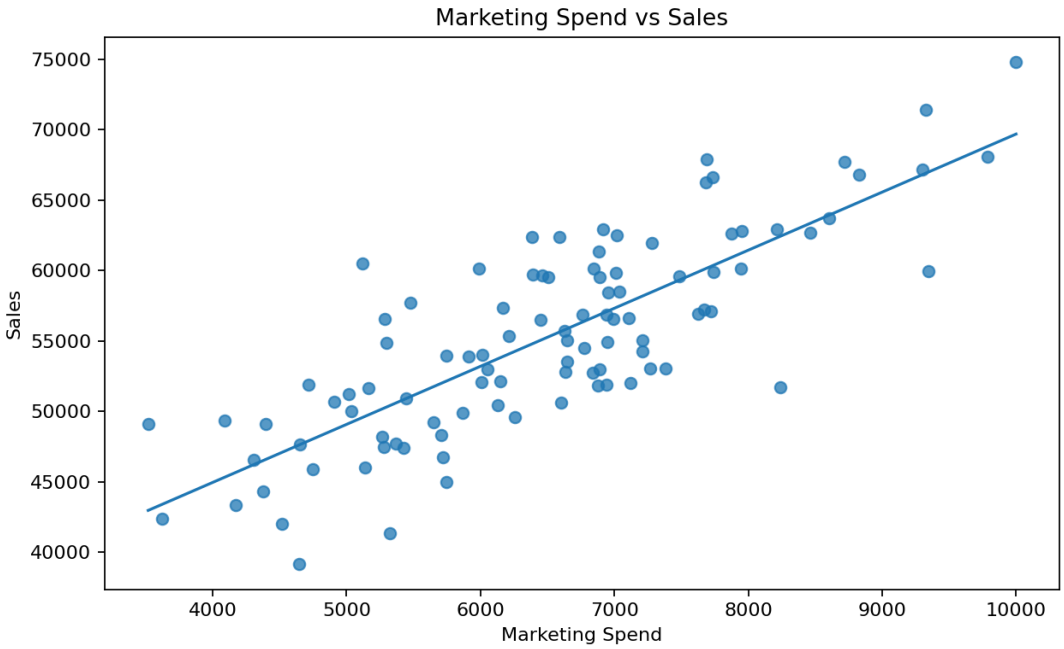
Mean sales = $55,257.63. The 95% confidence interval is $53,871.09 to $56,644.17, with a margin of error of ±$1,386.54.

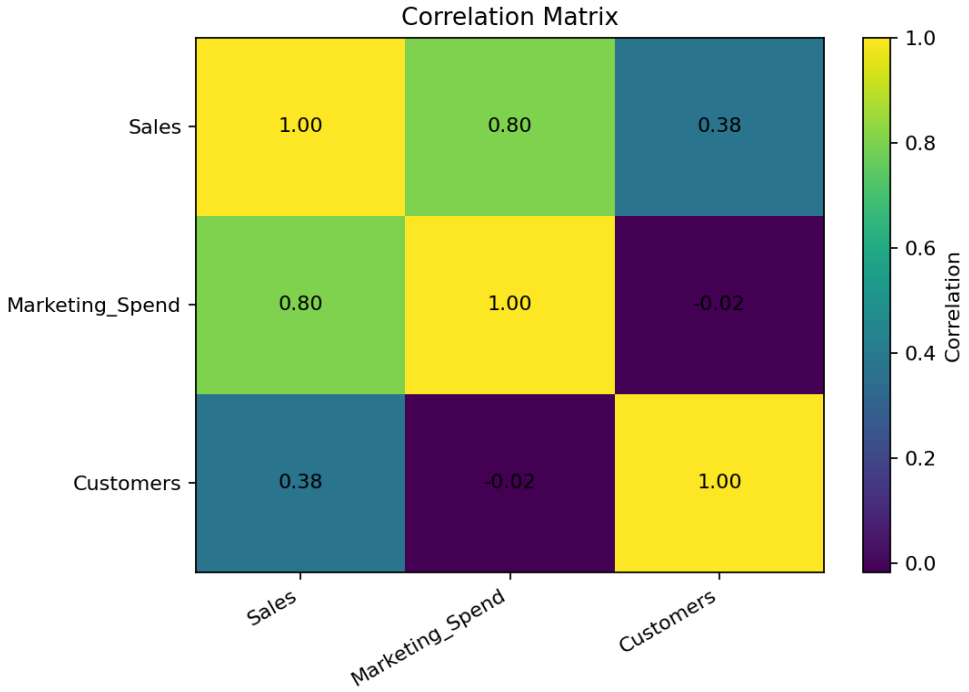
Regression Analysis

The simple linear regression predicts Sales from Marketing Spend. Intercept = 28,482.99; coefficient = 4.1214; R-squared = 0.6474; RMSE = 4,128.59. R-squared indicates the proportion of sales variation explained by marketing spend in this model.

Visual Documentation







Business Insights and Recommendations

• Use statistical significance rather than averages alone when evaluating business changes.
• Monitor regional performance where tests indicate meaningful differences.
• Treat marketing spend as an important explanatory variable while recognizing that correlation/regression does not by itself prove causation.
• Use the confidence interval to communicate uncertainty around average sales.
• Combine statistical evidence with operational context before making budget or sales decisions.

Technical Requirement Compliance

All required technical elements are implemented: descriptive statistics, three hypothesis tests, correlation analysis, confidence intervals, linear regression, R-squared, distribution plots, and business interpretation.

Testing Evidence

The dataset was checked for shape, missing values, duplicate rows, and successful completion of each required statistical component. Detailed evidence is included in screenshots/04_testing_evidence.txt.